

Datasets Overview

- Planting Date Trial
 - 1800 observations
 - 22 environments
- Varieties Test
 - 6700 observations
 - 586 environments
- 50 Predictors:
 - Planting Date
 - Maturity Group
 - Pop / RS
 - Soil and Weather

XGBoost

- Hyperparameters

Hyperparameter	Tested range	Interpretation
Learning rate	0.01 – 0.30	Controls the step size of each update
Number of trees	1000 – 5000	Number of boosting rounds
Tree depth	1 – 20	Maximum complexity of each tree
Subsample	0.50 – 1.00	Proportion of observations used per tree
mtry	1 – 15	Number of predictors evaluated at each split
Loss reduction	1e-10 – 31.6	Minimum loss reduction required to create a split
min_n	2 – 40	Minimum number of observations in a terminal node

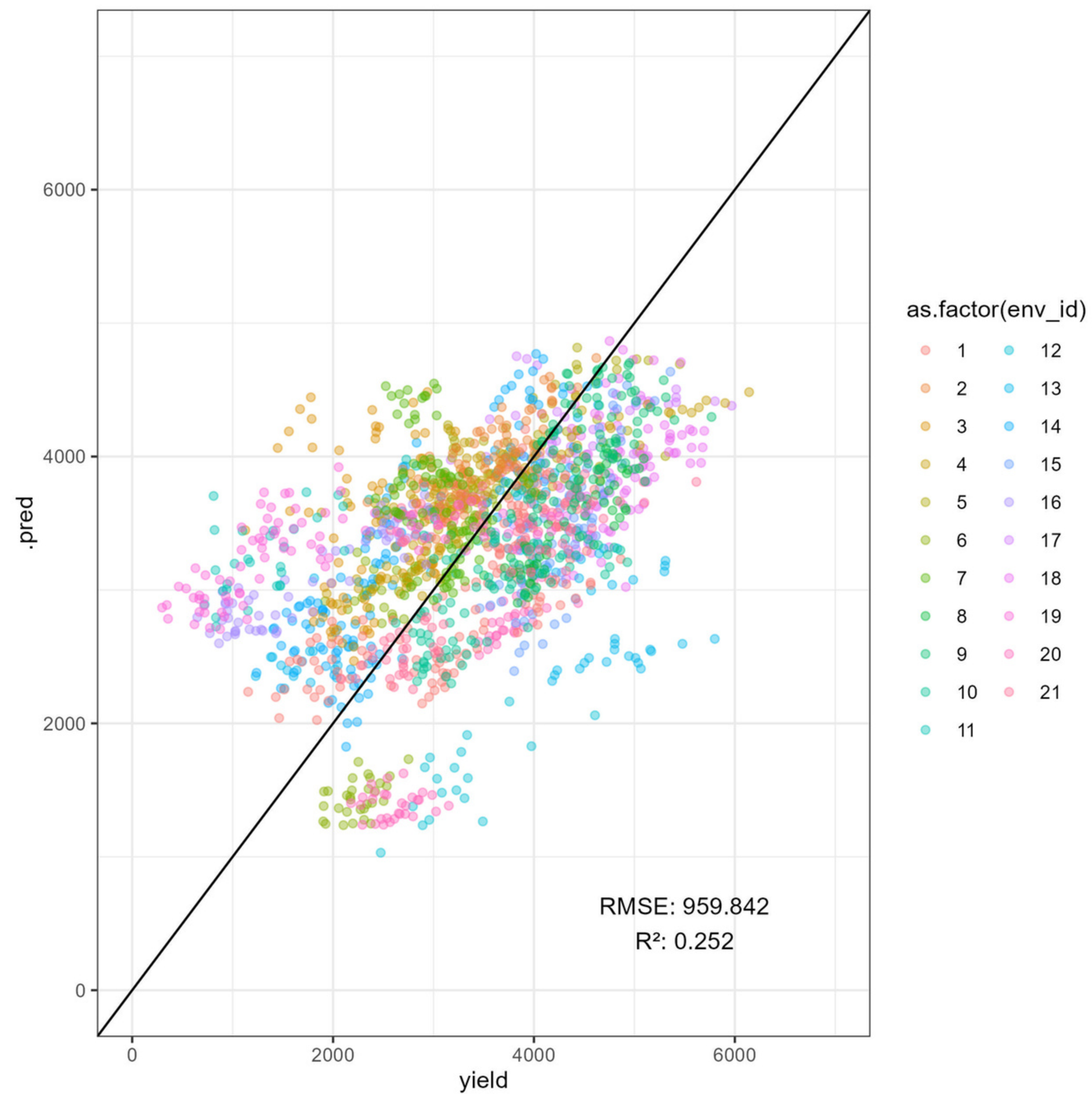
- Tuning (tune_bayes())
 - Initial candidates: 12
 - Iterations: 25
 - No improvement: 10

Nested Cross-Validation

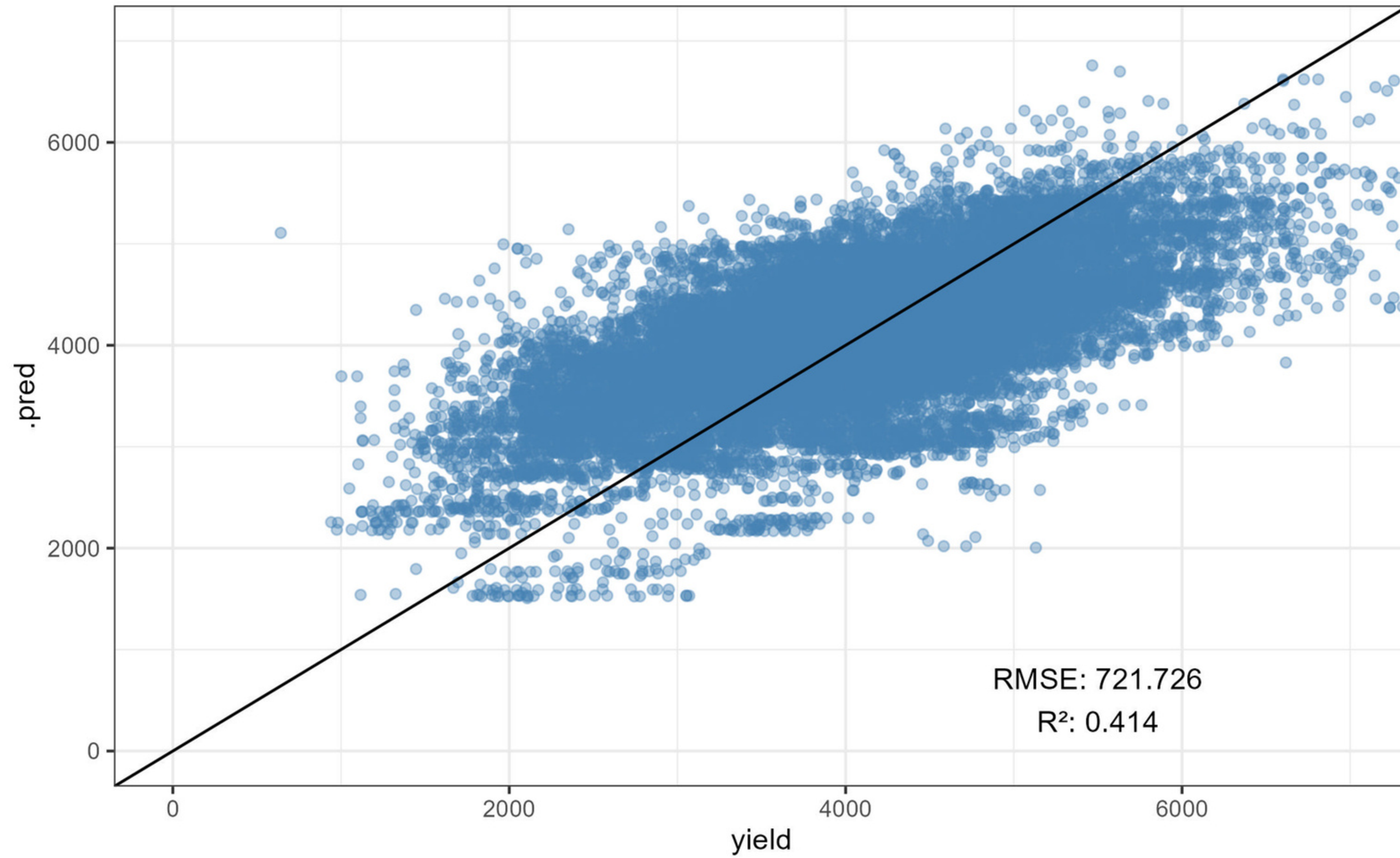
- Outer folds: 10, grouped by env_id
- Inner folds: 5, grouped by env_id
- Inner folds: used to tune hyperparameters
- Outer folds: used to estimate performance

Results

Planting Date Performance



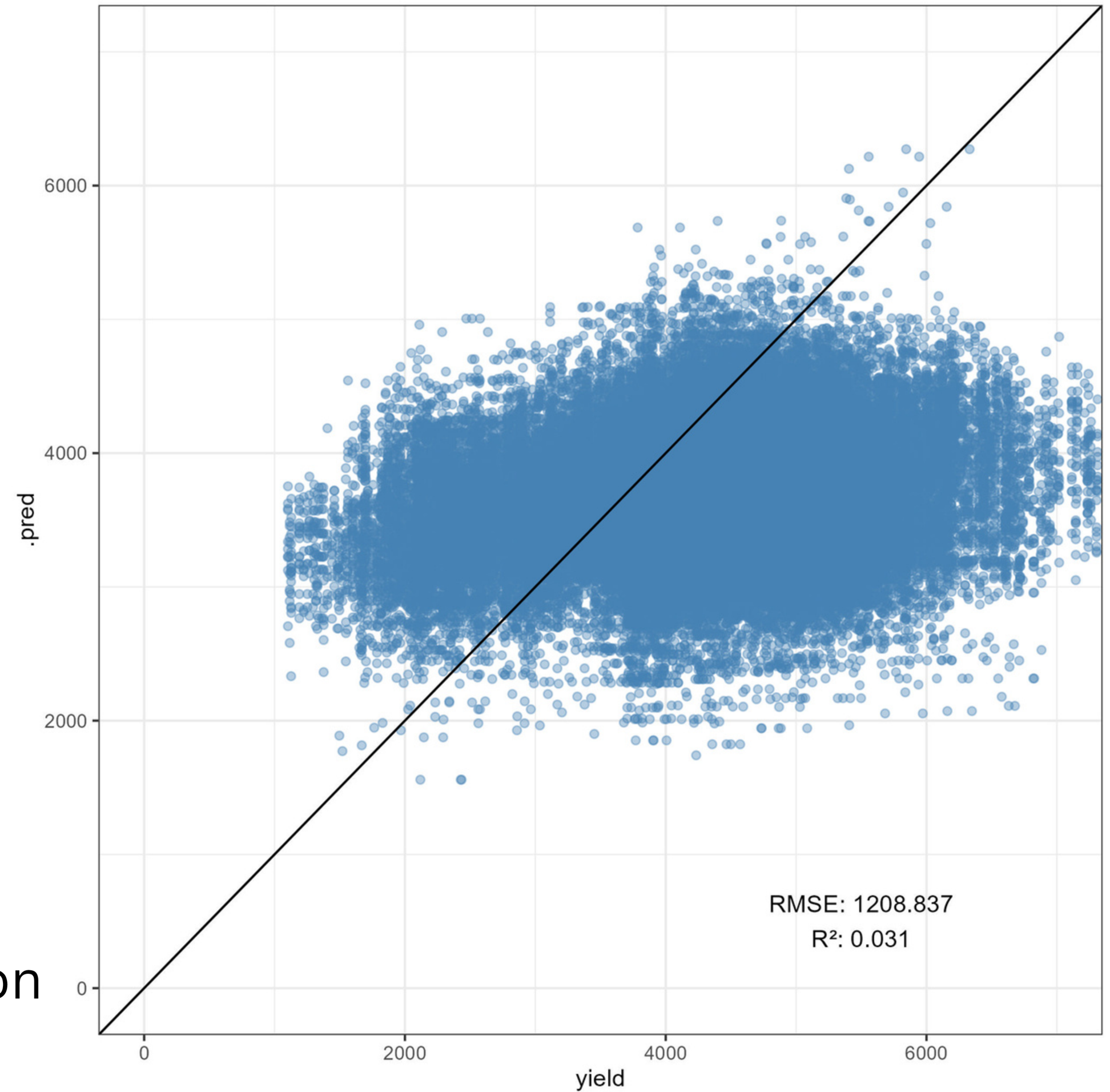
Results



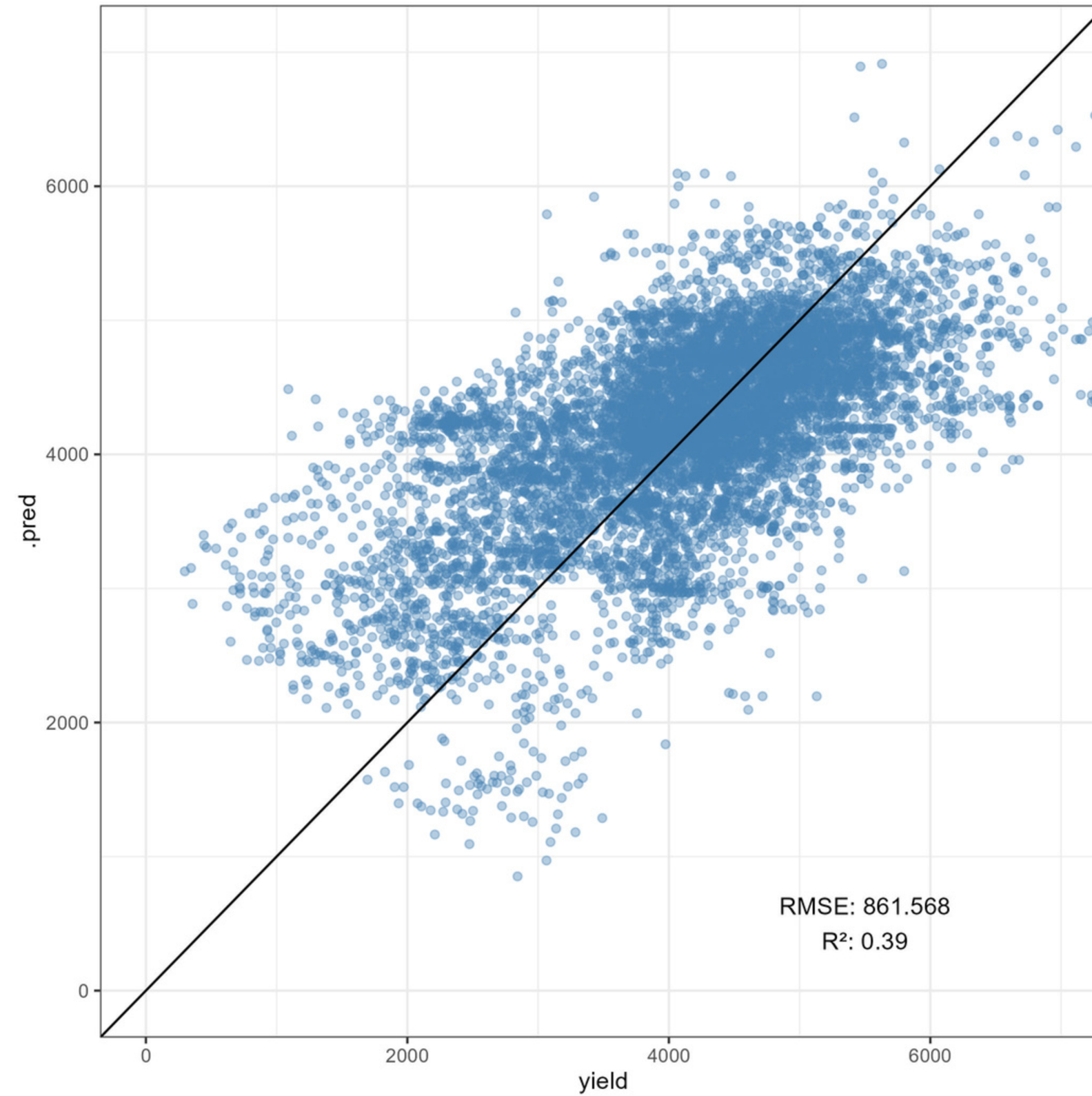
Varieties Test Performance

Results

Tuned on Planting Date Data, Tested on
VT Data



Results



Planting Date and VT Data Combined into One Dataset